Analytics Capstone Project (ANCP)

Sparta: Wardrobe Demand Analytics for Brands

Deliverable D2: Data audit report and datasheet

Group number	M
Deliverable code / version	D2 / v1
Date	3 November 2026
Lead owner for this deliverable	C. Prashanth
Term	Term 5, 6 October to 20 November 2026
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Linked outcomes	CO2, CO6
Dataset version audited	data-v1.0 (cleaned H&M 10% customer sample)

Data audit report

1. Purpose and scope

Sparta is a free digital wardrobe for men. With each user's consent, anonymised and aggregated wardrobe, purchase and wishlist data goes to brands and retailers to support two decisions: which categories, colours and fits to add to a range (primary), and how much stock to commit by category (secondary). The app also gives each user rule-based outfit suggestions from their own wardrobe. The process flow in the project repository (`flow-diagrams/sparta_process_flow.png`) shows both sides.

The pilot has not produced data yet. This audit therefore covers the data the team uses before the pilot: the public H&M Personalized Fashion Recommendations files, which stand in for purchase behaviour and seasonal demand (charter D1, Table 4). It states what the files contain, what we changed, and where they cannot stand in for Sparta's own users. The same audit will be repeated on the first Sparta pilot extract.

Source	Role in the project	Status in this audit
H&M articles, customers and transactions	Proxy for purchase behaviour and seasonality; checks the demand and gap methods before pilot data exists	Audited and cleaned (sections 2 to 6)
Sparta pilot app data (primary source)	Items, purchases, wishlist and outfits entered by consenting users	Not yet available; audit repeated on the first extract; consent rules in section 3
Synthetic records in the Sparta schema	Pipeline and script tests only	Used only to test `clean_hm.py`; never used for results

Table 1. Data sources and their status. Google Trends, listed as optional in the charter, is not used.

2. Source and access

Item	Detail
Dataset	H&M Personalized Fashion Recommendations (H&M Group, 2022): `articles.csv` (105,542 rows, 25 columns), `customers.csv` (1,371,980 rows, 7 columns) and the transactions file (31.8 million rows, 5 columns: date, customer, article, price, sales channel; 20 September 2018 to 22 September 2020).
Where from	Kaggle competition page, downloaded with a team member's Kaggle account after accepting the competition rules.
Licence or permission	Competition rules: use for non-commercial research and education. The team uses it only for this course and does not redistribute the raw files.
Date accessed	October 2026, before 27 October 2026, the date of the first audit run on the files.
Working sample	`transactions_train_sample10pct.csv`: all purchases of a random 10% of customers, 3,205,663 rows from 136,670 customers. A reference profile of the full 31.8 million rows (department, channel and monthly counts, purchases per customer, basket size) is kept so the sample can be tested against the whole file (section 5).
Where it is kept	Team members' machines only. Raw and cleaned row-level files are not committed to the public Sparta repository (size and licence). Each run records SHA-256 checksums of inputs and outputs in `MANIFEST.json` (section 7).

Table 2. Source and access record for the H&M data.

3. Consent and purpose limitation (DPDP Act, 2023)

H&M data. The files carry no names or contact details. Customer identifiers and postal codes are hashed by the publisher. We use the data only for this course (non-commercial research and

education), which is the purpose the competition rules allow. We do not try to identify anyone, and we never link H&M customers to Sparta users. We apply data minimisation: the postal-code column is never loaded, free-text product descriptions are not used, and only seven article columns and six customer columns are read. Customers under 18 are removed before analysis. Raw files stay on team machines and are deleted when the course ends; only aggregated results are kept.

Sparta pilot data. The Digital Personal Data Protection Act, 2023 (Government of India, 2023) expects a clear notice of what is collected and why, and consent that is free and specific (sections 5 and 6), and it sets stricter rules for children's data (section 9). Sparta's design follows these points: invitation-only adults in supported countries, a first-login pilot notice that every user must accept to join the study, a *Leave the pilot* button that deletes the account and data, minimum group sizes of 10 before any figure reaches a vendor, and no use of participant data to train machine-learning models (business requirement BR-28). The wording is in the Privacy Policy and the legal pack in the project repository. This statement is the team's own reading of the Act. No professional legal review has taken place, and one is required before any public or commercial launch.

4. Quality audit and treatment

Missing values, duplicates and outliers were checked on the raw files before any change. Table 3 shows each finding, the treatment chosen and the reason. The same rules run in `clean_hm.py` and in the D3 notebook.

#	Finding (raw data)	Treatment	Why
1	296,434 exact duplicate transaction rows (9.2% of the sample): same customer, day, article, price and channel. The file has no quantity column.	Keep. Collapse to one line with a quantity, so units are preserved.	Identical rows are most likely two units bought together, so dropping them would understate basket size.
2	Baby/Children department: 3.8% of units.	Exclude.	The buyer is dressing someone else, so the purchase says nothing about their own wardrobe.
3	Non-apparel product groups (cosmetics, furniture, stationery, "Unknown" and similar): 0.3% of units.	Exclude.	Not part of an outfit.
4	913 sampled customers under 18 (9,553 of all 1,371,980 customers, aged 16 or 17).	Exclude their purchases.	Sparta serves adults only.
5	Age above 90: 5 sampled customers (77 overall; maximum 99).	Set age to missing, keep the customer.	Likely default values, not real ages.
6	Missing demographics, whole customer file: FN flag 65.2%, Active flag 66.2%, age 1.2% (15,861), club status 0.4%, newsletter frequency 1.2%.	Age: explicit "unknown" band. FN and Active: missing means 0. Club status and newsletter: "UNKNOWN".	Imputing age would invent a signal. Missingness itself is informative (section 5).
7	Price is on a normalised scale (0.00003 to 0.507), not a currency. No zero, negative or missing prices; no missing values in the transactions.	Use only as a relative price. No outlier removal.	No value is impossible, so removal would hide real behaviour. Currency claims would be unfounded.
8	First and last weeks of the window are partial.	Drop them from weekly analyses only.	Avoids false dips at the window edges.
9	`sales_channel_id` is an anonymous code (1 or 2). It splits 29.6% / 70.4%.	Keep as a code; never label it "store" or "online".	The data dictionary does not say which is which.

10	Articles: 105,542 unique 10-digit ids, no duplicates; only `detail_desc` has gaps (416, 0.4%). Customers: no duplicate ids. One postal-code hash is shared by 120,303 customers (8.8%), probably a placeholder for a missing value.	Keep ids as text so leading zeros survive. Do not use postal code.	Reading ids as numbers breaks joins. The shared hash carries no location information.

Table 3. Quality findings, treatment and reasons. Counts come from the 10% sample unless "overall" or "whole file" is stated.

Step	Units	Share of raw
Raw rows (one row is one unit)	3,205,663	100.0%
Lines after collapsing exact duplicates (decision 1)	2,909,229 lines	n/a
After dropping Baby/Children (decision 2)	3,082,273	96.2%
After dropping non-apparel groups (decision 3)	3,071,946	95.8%
After dropping under-18 customers (decision 4): cleaned dataset	3,066,344	95.7%

Table 4. Cleaning funnel. The cleaned dataset holds 132,435 customers; Menswear is 5.8% of its units.

5. Representativeness and bias checks

Is the 10% sample like the whole file? Yes, on every check, on raw and on cleaned data (Table 5, Figure 1). The standardised mean difference (SMD) is the gap in means divided by the spread; below 0.1 is the usual sign of balance (Austin, 2009).

Check	Raw	Cleaned	Tolerance	Result
Customer age and club status (largest SMD)	0.003	0.021	below 0.1	Pass
Department mix (largest gap, percentage points)	0.052	0.046	below 1	Pass
Channel mix (largest gap, pp)	0.019	n/a	below 1	Pass
Monthly volume profile (largest gap, pp)	0.060	0.050	below 1	Pass
Purchases per customer, mean (gap)	0.5%	n/a	below 5%	Pass
Single-item visit share (gap, pp)	0.054	n/a	below 1	Pass

Table 5. Sample against the full 31.8 million-row file. Source: D3 notebook, section 3.2.

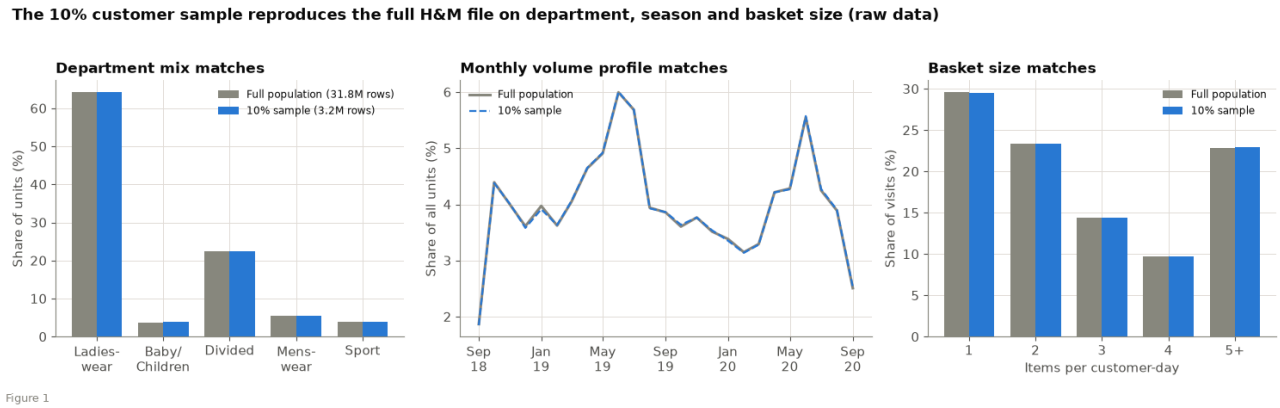


Figure 1. The 10% sample reproduces the full file on department mix, monthly volume and basket size (raw data). Reproduced from the D3 notebook.

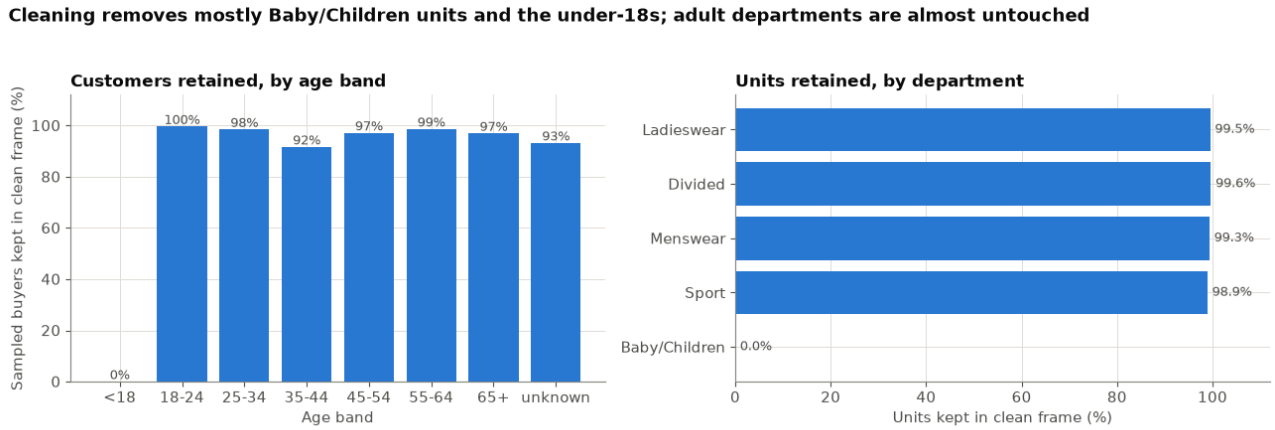


Figure 2

Figure 2. What our cleaning filters change: mostly Baby/Children units and under-18 customers; adult departments are almost untouched. Reproduced from the D3 notebook.

Who or what is under-represented, for Sparta's purposes? The sample matches H&M, but H&M is not Sparta. Table 6 lists the gaps that matter and what we do about each.

Gap or bias	Evidence	Effect on Sparta	Mitigation
Men are few and cannot be identified directly	No gender field. Menswear is 5.6% of raw units (5.8% of cleaned). Department is the only proxy.	Men's behaviour is thinly covered and may include gifts or shared accounts.	Report Menswear separately (`is_menswear` flag); use department as a labelled proxy; re-test on pilot data.
European customers and assortment	Publisher is European; prices normalised.	India's price ladder, ethnic wear and climate are absent.	Treat results as patterns to re-validate on pilot data; no rupee or euro claims.
Purchases only	No browsing, returns, ownership or wishlist.	A purchase is a weak signal of liking; ownership is invisible.	Ownership and wishlist come from the Sparta app; H&M tests only the purchase side.
Light buyers have missing demographics	Unknown age: median 1 visit against 3; FN missing: 2 against 4; unknown or pre-create club status: 1 against 3 (all p<0.001).	Missing values mark a different kind of customer.	Keep an explicit "unknown" level instead of imputing; report by activity tier.
Heavy buyers dominate	Top 10% of customers buy 48% of units (Gini 0.64); 32% of customers have a single visit.	Training data is shaped by a minority.	Split by customer; weight or report by activity tier.
A break in 2020	Mean weekly units fell 13.1% in March to August 2020 against 2019; channel 2 rose from 69.8% to 75.1% of units.	A pooled model could learn the break as a pattern.	Validate by date; compare with and without 2020.
Age mix is young	Customers aged 18 to 34 are 54% of all customers; 65 and over are 3%.	Older men are under-represented.	Report results by age band; flag thin bands.

Table 6. Under-representation and bias, with mitigations.

6. Feature engineering log

Table 7 lists each derived field: its logic and its leakage risk. Features 1 to 8 are written to the cleaned dataset by `clean_hm.py`. Features 9 to 14 are built in the D3 notebook and are not stored. Features 15 to 17 are planned for the Sparta pilot data and are listed so the same checks apply later.

#	Feature	Logic	Leakage risk and control
1	`qty`	Number of identical rows collapsed into one line	Low. Describes the same-day purchase only.
2	`role`	Product group mapped to Top, Bottom, Full-body, Accessory, Footwear, Hosiery, Underwear, Nightwear or Swimwear	None. Fixed catalogue attribute.
3	`is_menswear`	Index group equals Menswear	None. A proxy, not a gender label.
4	`age_clean`, `age_band`	Age above 90 set to missing; bands under 18, 18 to 24, 25 to 34, 35 to 44, 45 to 54, 55 to 64, 65 and over, unknown	Low. A customer attribute, constant over time.
5	`FN`, `Active`	Missing filled with 0	Low. Missing meaning 0 is an assumption, tested in section 5.
6	`club_member_status`, `fashion_news_frequency`	Missing filled with UNKNOWN; "None" merged with NONE	Medium. Status is recorded at one point in time, so it may reflect behaviour that comes after a purchase.
7	`week_start`	Monday of the purchase week	None. Calendar field.
8	`is_minor` (used to filter, not stored)	Age under 18; missing age is not a minor	None.
9	Visit	One customer on one day	None.
10	Single-unit visit, top-and-bottom visit	Visit has one unit; visit has a Top and a Bottom role	Low. Same-day content only.
11	History hit	Customer bought the same article, type, colour or department on an earlier day than the item	High if misused. Built only from days before the item; the baseline shuffles within the same week; models must split by customer and date.
12	Adjusted lift of a type pair	Visits containing both types divided by the number expected when items are shuffled among visits of the same size and week	Medium. Uses the whole window; fine for description, not for evaluation.
13	Top-100 weekly overlap, new-article share	Jaccard overlap of the top 100 articles at 1, 4, 13 and 52 weeks; share of units from articles first sold up to 8 weeks earlier	Medium. Uses future weeks; any forecast must be fit on earlier weeks only.
14	Customer activity measures	Visits, units, single-unit share, distinct product types per customer	High. Computed over the whole window, so they must be rebuilt inside the training window for modelling.
15	Colour family (Sparta)	Nearest reference swatch in CIELAB, from the colour the user picks	None. Derived from the item alone.
16	Condition band (Sparta)	Condition 0 to 100 grouped into 0 to 9, 10 to 29, 30 to 59, 60 to 84, 85 to 100	Low. User-entered, so it may be inconsistent between users.
17	Gap score (Sparta)	Distinct users with an open wishlist entry in a segment divided by distinct users who own an item in it	Medium. Wishlist and ownership are measured at the run date; the holdout is the last 10 days of activity and is never used in tuning.

Table 7. Feature engineering log.

7. Reproducibility, versioning and limits

Rerun. `submissions/D2/clean_hm.py` rebuilds the cleaned dataset from the three raw files: `python clean_hm.py --raw-dir <raw> --out-dir <out>`. Rows are sorted and files are written without timestamps, so the same input gives byte-identical output. It writes `lines_clean.csv.gz`, `customers_clean.csv.gz` and `articles_clean.csv.gz` plus `MANIFEST.json`, which records the SHA-256 checksum, row count and columns of every input and output, the cleaning funnel and the library versions. The data dictionary is `data_dictionary.csv` (Appendix B). The release is tagged data-v1.0 in the team repository once the run on the real files has produced its checksums.

Checked so far. The script was run twice on a synthetic file with the real column layout: both runs gave identical checksums and the funnel followed the rules in Table 3. The numbers in this report come from the team's run on the real files, shown in the D3 notebook and the audit statistics file. The script itself has not yet been run on the real files; that run produces the checksums for data-v1.0.

Limits. The data show purchases by European shoppers in 2018 to 2020. They cannot show ownership, wishlist demand or Indian menswear. They support method checks (seasonality, concentration, cold start) but not conclusions about Sparta's users. The pilot extract replaces them as soon as it exists.

References

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Appendix A. Datasheet: H&M cleaned sample (data-v1.0)

Structure follows Gebru et al. (2021). The datasheet describes the cleaned dataset the team derives from the public H&M files, not the Sparta pilot data.

A1. Motivation

- **Purpose.** Give Sparta a realistic purchase history to test its demand, seasonality and gap-ranking methods before the pilot produces data. It was not made for Sparta by H&M.
- **Who created the source.** H&M Group released the original files for a public recommendation competition (2022). The team created the cleaned dataset for course work.
- **Funding.** None. A student project.

A2. Composition

- **Instances.** Cleaned transaction lines: one line per customer, day, article, price and channel, with a quantity. The cleaned frame holds 3,066,344 units from 132,435 customers (Table 4). Two companion tables describe customers and articles.
- **Source scale.** Whole customer file 1,371,980 rows; whole article file 105,542 rows; the transaction file is a 10% customer sample of 31.8 million rows, 20 September 2018 to 22 September 2020.
- **Sample.** All purchases of a random 10% of customers. It reproduces the full file on every check in Table 5.
- **Fields.** Date, week start, hashed customer id, article id, price (normalised), sales channel code (1 or 2), quantity, article role and attributes (product type, product group, garment group, colour, department), customer age band and flags. See Appendix B.
- **Labels or targets.** None. Targets are defined in later deliverables.
- **Missing data.** Customer demographics (Table 3, row 6). Transactions and article ids are complete.
- **Relationships between instances.** Lines of one customer are linked by the hashed id; lines of one day form a visit.
- **Splits.** None supplied. Modelling splits are made by customer and by date (D4).
- **Noise and errors.** Exact duplicate rows (kept as quantity), implausible ages above 90 (set to missing), an anonymous channel code, and a shared postal-code hash (not used).
- **Confidential or sensitive content.** No names or contact details. Customer ids are publisher-made hashes, and age is held as a band in the cleaned file. The postal code is not loaded.
- **Does it identify people?** Not directly; re-identification from purchase patterns is conceivable, so we never attempt it and never link the data to Sparta users.

Composition of the source files (whole files, before cleaning)

Department (articles file)	Articles	Share	Kept in the cleaned dataset
Ladieswear	39,737	37.7%	Yes
Baby/Children	34,711	32.9%	No (decision 2)
Divided	15,149	14.4%	Yes
Menswear	12,553	11.9%	Yes, flagged `is_menswear`
Sport	3,392	3.2%	Yes

Table 8. Articles by department. The 105,542 articles fall in 8 main product groups: upper-body garments (42,741), lower-body (19,812), full-body (13,292), accessories (11,158), underwear (5,490), shoes (5,283), swimwear (3,127) and socks and tights (2,442), plus smaller groups. The most common catalogue colours are Black (22,585), Blue (18,469), White (12,665), Pink (9,403) and Grey (8,924).

Age band (customers file)	Customers	Share	Age band	Customers	Share
Under 18 (16 and 17)	9,553	0.7%	45 to 54	253,951	18.5%
18 to 24	347,616	25.3%	55 to 64	139,064	10.1%
25 to 34	393,292	28.7%	65 and over	43,602	3.2%
35 to 44	169,041	12.3%	Age missing	15,861	1.2%

Table 9. Customers by age band, all 1,371,980 customers. Median age 32, mean 36.4, range 16 to 99.

A3. Collection process

- **How the data was acquired.** H&M recorded sales to club members and online and store buyers; the publisher hashed identifiers and normalised prices before release. The team did not collect it.
- **Sampling.** The team drew all purchases of a random 10% of customers from the full transaction file.
- **Time frame.** 20 September 2018 to 22 September 2020. The record includes the 2020 disruption (Table 6).
- **Consent.** The data subjects' consent is the publisher's responsibility; we rely on the competition's terms for non-commercial research and education and do not process the data for any other purpose (section 3).
- **Ethical review.** No formal review; the team's privacy assessment covers the Sparta pilot, not this dataset.

A4. Preprocessing, cleaning and labelling

- **What was done.** Ten checks and decisions (Table 3); collapse duplicates to quantity; drop Baby/Children, non-apparel and under-18 purchases; set ages above 90 to missing; fill missing flags; add role, `is_menswear`, age band and week start (Table 7).
- **Raw data.** Kept unchanged on team machines, outside the repository.
- **Software.** `clean_hm.py` (Python, pandas, NumPy); the run writes `MANIFEST.json` with checksums and library versions.

A5. Uses

- **Intended.** Descriptive analysis (D3), baselines and candidate models for seasonal demand and segment ranking (D4), and the checks on those models (D5), all as a proxy for the Sparta pilot.

- **Not suitable for.** Conclusions about Indian men's wardrobes, ownership or wishlist demand, per-person profiling, gender-specific claims, or any commercial use. Any score computed on it should be reported as a result on H&M data.
- **Risks to people.** Low, given pseudonymisation and minimisation, but the data describes real purchases; treat it as personal data.

Suitability for tasks

Task	Suitable?	Why
Seasonal demand patterns, concentration of sales, cold-start behaviour	Yes	Two full years of weekly data for 105,542 articles
Testing the gap-ranking and forecasting code and the validation design	Yes	A method check; every score stays labelled as a result on H&M data
Checking outfit rules against what is bought together	Partly	Co-purchase cannot separate complements from alternatives (D3 insight 5), so it can only suggest candidates
Ownership or wishlist demand	No	Not recorded; comes from the Sparta app
Claims about men, or about the Indian market	No	No gender field; a European assortment and customer base
Profiling individuals or any commercial use	No	Outside the competition rules and the purpose in section 3

Table 10. Suitability of the cleaned dataset for tasks.

A6. Distribution

- **Who gets it.** The course team and faculty, through the course submission folder and the team repository. It is not published, and not placed in the public Sparta repository.
- **Licence.** The derived files remain subject to the competition rules (non-commercial research and education). The code is GPL-3.0.

A7. Maintenance

- **Owner.** C. Prashanth (lead for D2), contactable through the team.
- **Updates.** The dataset is frozen at data-v1.0. Any change to the cleaning rules produces a new version tag and a new `MANIFEST.json`.
- **Retirement.** Raw and cleaned row-level files are deleted from team machines when the course ends; the report, the script and the aggregated results remain.

Appendix B. Data dictionary (cleaned dataset)

B1. `lines_clean.csv.gz` (one line per customer, day, article, price and channel)

Column	Type	Description
t_dat	date	Purchase date (YYYY-MM-DD)
week_start	date	Monday of the purchase week
customer_id	text	64-character hash from the publisher; joins to `customers_clean`
article_id	text	10-digit article id with leading zeros; joins to `articles_clean`
price	number	Normalised unit price; relative use only, not a currency
sales_channel_id	integer	Anonymous channel code, 1 or 2
qty	integer	Units in this line (identical rows collapsed)
role	text	Top, Bottom, Full-body, Accessory, Footwear, Hosiery, Underwear, Nightwear, Swimwear
is_menswear	boolean	True when the article's index group is Menswear (a proxy, not a gender label)
index_group_name	text	Department: Ladieswear, Divided, Menswear or Sport
product_group_name, product_type_name, garment_group_name	text	Catalogue hierarchy
perceived_colour_master_name, perceived_colour_value_name	text	Catalogue colour and its shade (for example Dark)
age_band	text	18-24, 25-34, 35-44, 45-54, 55-64, 65+ or unknown

Table 11. Columns of the lines file.

B2. `customers_clean.csv.gz` and `articles_clean.csv.gz`

File and column	Type	Description
customers: customer_id	text	As above; one row per customer who remains in scope
customers: FN, Active	0 or 1	Publisher flags; missing is stored as 0
customers: club_member_status	text	ACTIVE, PRE-CREATE, LEFT CLUB or UNKNOWN
customers: fashion_news_frequency	text	NONE, Regularly, Monthly or UNKNOWN
customers: age_clean, age_band	number, text	Age with values above 90 set to missing; band as above
articles: article_id and the columns above	text	One row per article that appears in the lines file; also holds `role` and `is_menswear`

Table 12. Columns of the customer and article files. The postal code, product name and description are not carried over.

Appendix C. AI-tool disclosure

The team used Claude (Anthropic) to draft this report's text, to write the cleaning script and to build the figures' layout. The team checked that every number traces to the D3 notebook outputs or the audit statistics, that the cleaning rules match the notebook, and that the script runs twice with identical checksums on synthetic data. The team did not use AI tools to produce any result from the real data.